* PS I
HCL
IBM
↓
Patient with higher priority gets Spented first.

* PS II
Infosys
Capgemini
↓
(Custom Comparator)
Because of multiple comparisons

1 → 3 → 5
↓ ↓ ↓
1st 2nd 3rd
Min Heap

The Trie Data Structure (Suffix Tree)

* Autocomplete in browsers
* Dictionary Search words
* LCP → Longest Common Prefix
* Trie based prefix search
* Phone Contact List

Why do we need Tries?

S1 ARYA ✓ <S1, true>
S2 ARYAN ✓ <S2, true>
S3 ARYAMAAN ✓ <S3, true>
S4 ARMAN ✓ <S4, true>
S5 AMAN ✓ <S5, true>

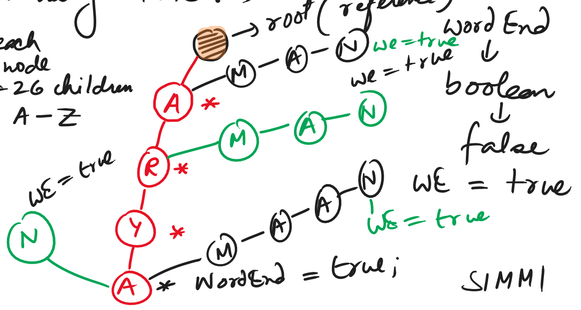
Suppose we need to store these names in a collection using some data structure, we can use a HashMap

HashMap<String, Boolean> hm = new HashMap<>();
<SAURAV, false> → doesn't exist
For every string extra space is guaranteed. (Space)

Same data stored using Trie: →

S1 ARYA ✓ →
S2 ARYAN ✓
S3 ARYAMAAN ✓
S4 ARMAN ✓
S5 AMAN ✓

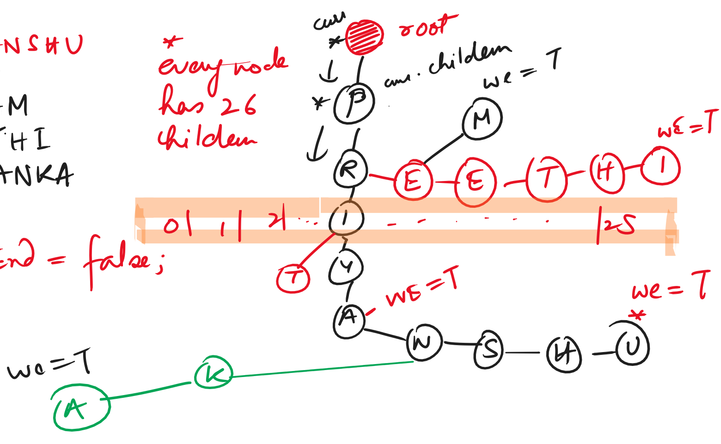
each node = 26 children A–Z



SIMMI

* We traverse each string character by character
* If node not present, create, if present traverse below it.

PRIYANSHU
PRIYA
PRITAM
PREETHI
PRIYANKA
PREM

* every node has 26 children

wordEnd = false;